AI Report

1.Prompts to ChatGPT & Responses Related to EDA

Prompt 1: Perform Exploratory Data Analysis (EDA) on the data.

Chatgpt Response:

This dataset is a relational database describing customers, agents, entertainers, musical preferences, and entertainment engagements. It resembles a small business system for booking entertainers.

1) Entity overview:

Table	Description
Customers	Customer demographics
Agents	Booking agents, salary & commission info
Entertainers	Performers, their info & styles
Engagements	Actual bookings (date, time, price, customer, agent, entertainer)
Members	Individual band members
Entertainer_Members	Link table: entertainer ↔ members
Entertainer_Styles / Musical_Styles	Various music styles
Musical_Preferences	Customer music preferences

This is a multi-table relational system with many-to-many relationships.

2) EDA on Key Tables

Customers

Count: The inserted IDs range from 10001 to 10015, so ≈15 customers.

Cities: Bellevue (WA), Seattle (WA), Kirkland (WA), Tacoma (WA),Redmond (WA) Auburn (WA)

All customers are in Washington State.

Agents

Count: Agents include IDs from 1 to 9 (≈9 agents)

Salary Distribution (from inserts):

Range: $50 to $35,000

One “dummy” record: salary = $50, commission = 1% (likely fake/test agent)

Commission Rates: Range: 3.5% – 6%

Entertainers

Count: Entertainer IDs 1001–1013 → ≈13 entertainers

Attributes: All located in WA

Some have web pages/emails, some missing → missing-value pattern

Style Information (partial because file truncated):

Stored in: Entertainer_Styles, Musical_Styles

Could analyze: Popular styles, Strength distribution

3) Engagements (most important table)

The Engagements table contains the entertainment bookings. This is the core for revenue EDA.

Variables:

Start & End Date

Start & Stop Time

ContractPrice

CustomerID

AgentID

EntertainerID

Number of Engagements: From visible inserts, EngagementNumber ranges from 2 → 131.Not continuous, but roughly ≥ 100 bookings.

Contract Price (Revenue)

Visible prices range:

- Lowest: $110
- Highest: $14,105
- Typical range appears: $200–$2,500

High-value bookings:

Examples:

- $14,105 (Engagement #99)
- $3,800
- $3,650

Skewed distribution — few large bookings, many mid-range.

Date Patterns

Engagements mostly from:

- Sep 2017 → Mar 2018

Suggests:

- Seasonal trend (holidays, year-end events)
- Bookings cluster around Thanksgiving, Christmas, and New Year

Time-of-day Patterns

Start times range: 12:00–20:00

Stop times are often “00:00” or “01:00” → entertainment events typically evening/night.

4) Business-Level EDA Questions

· Which agents generate the most revenue?

· Which entertainers are most booked?

- Customer booking behavior
- Seasonality
- Duration of engagements

5) Data Quality Findings

Missing Values
Entertainer emails/web pages missing for some entries
Members table has a NULL gender for MemberID 125
Engagement numbers not contiguous (okay)
Outliers
Agent salary of $50 is unrealistic → clearly a dummy/test row
One very large contract: $14,105 (Engagement #99)

Prompt 2: Help me generate SQL queries for Business-Level EDA Questions.

Chatgpt Response:
Which agents generate the most revenue?
Business angle: Who are our top-performing agents by sales and by number of gigs?
SQL Query:

```
SELECT
    a.AgentID,
    a.AgtFirstName || ' ' || a.AgtLastName AS agent_name,
    COUNT(e.EngagementNumber)          AS num_gigs,
    SUM(e.ContractPrice)          AS total_revenue,
    AVG(e.ContractPrice)          AS avg_contract
FROM Engagements e
JOIN Agents a
  ON e.AgentID = a.AgentID
GROUP BY a.AgentID, agent_name
ORDER BY total_revenue DESC;
```

Result:

	agentid	agent_name	num_gigs	total_revenue	avg_contract
1	3	Carol Viescas	19	24,800	1,305.2631578947
2	6	John Kennedy	12	24,435	2,036.25
3	5	Marianne Wier	18	22,635	1,257.5
4	1	William Thompson	16	19,895	1,243.4375
5	4	Karen Smith	17	18,595	1,093.8235294118
6	8	Maria Patterson	15	12,825	855
7	7	Caleb Viescas	8	10,645	1,330.625
8	2	Scott Bishop	6	6,720	1,120

Which entertainers are most popular and profitable?

Business angle: Who should we promote more / negotiate higher rates with?

SQL Query:

```
SELECT
    en.EntertainerID,
    en.EntStageName,
    COUNT(e.EngagementNumber) AS num_gigs,
    SUM(e.ContractPrice)      AS total_revenue,
    AVG(e.ContractPrice)      AS avg_contract
FROM Engagements e
JOIN Entertainers en
  ON e.EntertainerID = en.EntertainerID
GROUP BY en.EntertainerID, en.EntStageName
ORDER BY num_gigs DESC, total_revenue DESC;
```

Result:

	entertainerid	entstagename	num_gigs	total_revenue	avg_contract
1	1,008	Country Feeling	15	34,080	2,272
2	1,013	Caroline Coie Cuartet	11	15,070	1,370
3	1,001	Carol Peacock Trio	11	11,080	1,007.2727272727
4	1,003	JV & the Deep Six	10	17,150	1,715
5	1,006	Modern Dance	10	14,600	1,460
6	1,010	Saturday Revue	9	11,550	1,283.3333333333
7	1,004	Jim Glynn	9	3,030	336.6666666667
8	1,007	Coldwater Cattle Company	8	14,875	1,859.375
9	1,011	Julia Schnebly	8	4,345	543.125
10	1,002	Topazz	7	6,620	945.7142857143
11	1,005	Jazz Persuasion	7	5,480	782.8571428571
12	1,012	Susan McLain	6	2,670	445

Which customers are the most valuable?

Business angle: Identify VIP / key accounts.

SQL Query:

```
SELECT
    c.CustomerID,
    c.CustFirstName || ' ' || c.CustLastName AS customer_name,
    c.CustCity,
    COUNT(e.EngagementNumber)           AS num_bookings,
    SUM(e.ContractPrice)           AS total_spend,
    AVG(e.ContractPrice)           AS avg_contract
FROM Engagements e
JOIN Customers c
  ON e.CustomerID = c.CustomerID
GROUP BY c.CustomerID, customer_name, c.CustCity
```

ORDER BY total_spend DESC;

Result:

	customerid	customer_name	custcity	num_bookings	total_spend	avg_contract
1	10,005	Elizabeth Hallmark	Auburn	8	25,585	3,198.125
2	10,006	Matt Berg	Tacoma	9	13,170	1,463.3333333333
3	10,014	Mark Rosales	Bellevue	10	12,770	1,277
4	10,010	Zachary Ehrlich	Kirkland	13	12,455	958.0769230769
5	10,002	Deb Waldal	Tacoma	10	12,320	1,232
6	10,004	Dean McCrae	Redmond	11	11,800	1,072.7272727273
7	10,001	Doris Hartwig	Seattle	8	10,795	1,349.375
8	10,015	Carol Viescas	Seattle	7	8,255	1,179.2857142857
9	10,013	Estella Pundt	Bellevue	6	7,560	1,260
10	10,003	Peter Brehm	Kirkland	7	7,250	1,035.7142857143
11	10,009	Sarah Thompson	Bellevue	8	7,090	886.25
12	10,012	Kerry Patterson	Redmond	7	6,815	973.5714285714
13	10,007	Liz Keyser	Bellevue	7	4,685	669.2857142857

Is there seasonality in revenue (by month)?

Business angle: When is demand highest? (Budgeting/staffing.)

SQL Query:

```
SELECT
    DATE_TRUNC('month', e.StartDate) AS month,
    COUNT(e.EngagementNumber)        AS num_gigs,
    SUM(e.ContractPrice)             AS total_revenue,
    AVG(e.ContractPrice)             AS avg_contract
FROM Engagements e
GROUP BY DATE_TRUNC('month', e.StartDate)
ORDER BY month;
```

Result:

month	num_gigs	total_revenue	avg_contract
2017-09-01 00:00:00.000 -0700	16	20,135	1,258.4375
2017-10-01 00:00:00.000 -0700	22	30,125	1,369.3181818182
2017-11-01 00:00:00.000 -0700	7	5,600	800
2017-12-01 00:00:00.000 -0800	17	17,275	1,016.1764705882
2018-01-01 00:00:00.000 -0800	28	43,880	1,567.1428571429
2018-02-01 00:00:00.000 -0800	20	21,685	1,084.25
2018-03-01 00:00:00.000 -0800	1	1,850	1,850

How long are engagements? (Duration in days)

Business angle: Typical contract length; identify very long/short gigs.

SQL Query:

```
SELECT
    e.EngagementNumber,
```

```
    e.StartDate,
    e.EndDate,
    (e.EndDate - e.StartDate) AS duration_days,
    e.ContractPrice,
    c.CustFirstName || ' ' || c.CustLastName AS customer_name,
    en.EntStageName
FROM Engagements e
JOIN Customers c
  ON e.CustomerID = c.CustomerID
JOIN Entertainers en
  ON e.EntertainerID = en.EntertainerID
ORDER BY duration_days DESC, ContractPrice DESC;
```

Result:

	engagementnumber	startdate	enddate	duration_days	contractprice	customer_name	entstagename
1	99	2018-01-23	2018-02-23	31	14,105	Elizabeth Hallmark	Country Feeling
2	75	2018-01-02	2018-01-12	10	2,525	Doris Hartwig	Coldwater Cattle Compar
3	88	2018-01-08	2018-01-18	10	1,370	Dean McCrae	Caroline Coie Cuartet
4	9	2017-09-19	2017-09-29	10	1,370	Zachary Ehrlich	Saturday Revue
5	72	2017-12-23	2018-01-02	10	875	Kerry Patterson	Julia Schnebly
6	36	2017-10-14	2017-10-24	10	710	Mark Rosales	Julia Schnebly
7	28	2017-10-07	2017-10-16	9	3,800	Peter Brehm	Coldwater Cattle Compar
8	10	2017-09-18	2017-09-27	9	3,650	Elizabeth Hallmark	JV & the Deep Six
9	31	2017-10-08	2017-10-17	9	2,450	Deb Waldal	Saturday Revue

For a summary of duration distribution:
SQL Query:

```
SELECT
    AVG(e.EndDate - e.StartDate) AS avg_duration_days,
    MIN(e.EndDate - e.StartDate) AS min_duration_days,
    MAX(e.EndDate - e.StartDate) AS max_duration_days
FROM Engagements e;
```

Result:

	avg_duration_days	min_duration_days	max_duration_days
1	5.0900900901	0	31

Revenue by time of day

Business angle: When do we usually run events (afternoon vs evening)?
SQL Query:

```
SELECT
    DATE_PART('hour', e.StartTime) AS start_hour,
    COUNT(*)              AS num_gigs,
    SUM(e.ContractPrice)        AS total_revenue
```

FROM Engagements e
GROUP BY start_hour
ORDER BY start_hour;

Result:

	start_hour	num_gigs	total_revenue
1	12	13	19,445
2	13	13	13,850
3	14	15	32,130
4	15	10	10,925
5	16	12	12,795
6	17	16	20,450
7	18	9	6,960
8	19	6	5,775
9	20	17	18,220

Prompt 3:What business insights can we derive from this EDA?

Chatgpt Response:

1. Agent Performance Insight: Revenue is highly concentrated

Two agents — Carol Viescas and John Kennedy — contribute the majority of total revenue. Mid-tier agents generate stable but moderate income. Lower-performing agents have noticeably fewer bookings and lower revenue.

Business Implication:

High-performing agents should be prioritized for retention and incentive programs.Lower-performing agents may benefit from training, resource support, or redistribution of booking opportunities.

2. Entertainer Popularity Insight: A small set of entertainers drives most gigs

Country Feeling is the clear leader in booking volume.Several entertainers maintain moderate demand.A long tail of performers show relatively low engagement frequency.

Business Implication:

High-demand entertainers should receive premium pricing, targeted promotion, and priority scheduling.Under-booked entertainers may need marketing boosts, repositioning, or brand refresh strategies.

3. Customer Value Insight: A few VIP customers generate disproportionate revenue

Elizabeth Hallmark dramatically outspends all other customers. Several mid-tier customers provide consistent bookings. The rest form a long tail with smaller but stable spend.

Business Implication：High-value customers should be placed into a VIP loyalty program with personalized offers.Mid-tier customers represent a strong segment for upsell and cross-sell opportunities. Long-tail customers are ideal for automated marketing and email campaigns.

4. Seasonality Insight: Revenue peaks in January and dips in November

January shows the highest revenue, driven by New Year and post-holiday events. October and February are also strong months. November is the weakest month.

Business Implication:

The company can develop seasonal pricing strategies, such as peak-season premiums. November could be targeted for discount campaigns or special packages to fill demand gaps.

5. Engagement Duration Insight: Most bookings are short-term (2–8 days)

The distribution shows a clear peak around 2–8 days. Very long engagements are rare.

Business Implication:

Short-term booking models are the norm → staff and inventory planning should be flexible and agile. Long-term bookings could be encouraged through discounts or bundled pricing.

6. Time-of-Day Insight: Most engagements start in the afternoon/evening

Engagements cluster from 12:00 to 20:00, peaking around 20:00. This aligns with typical entertainment timing (corporate dinners, concerts, events).

Business Implication:

Resource allocation (equipment, staff, transportation) should be optimized for late afternoon/evening peak loads. Morning slots represent an opportunity for new product offerings such as workshops, brunch entertainment, or corporate morning events.

Overall Strategic Insight

This EDA reveals clear patterns in revenue concentration, demand cycles, and customer behavior: Revenue is driven by top agents, top entertainers, and a few VIP customers. Entertainment demand is seasonal and time-specific, peaking during evening hours and early-year celebrations. The business operates primarily in short-term engagements, suggesting flexibility is key.

2.Process Used to Create Data Visualizations

Prompt: Help me generate the python code to visualize the EDA results.

Setup: Load data into a notebook(Python) and load the SQL file into SQLite

Python code:

```
import pandas as pd
import matplotlib.pyplot as plt
import sqlite3
import re
# 1. Connect to an in-memory SQLite database
conn = sqlite3.connect(":memory:")
cur = conn.cursor()
# 2. Read your entertainment.sql file
sql_path = "/Users/tiffsy/Downloads/entertainment.sql"  # <<< change to your actual path
```

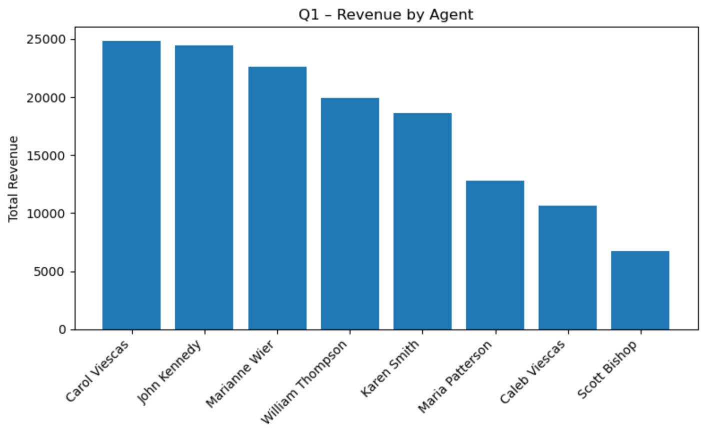
```
with open(sql_path, "r", encoding="utf-8") as f:
    sql_text = f.read()
# 3. Convert PostgreSQL-specific types to SQLite-friendly ones
sql_text = re.sub(r"\bserial\b", "INTEGER", sql_text)
sql_text = re.sub(r"decimal\([^)]+\)", "REAL", sql_text)
sql_text = re.sub(r"float\([^)]+\)", "REAL", sql_text)
# (date/time can stay as TEXT in SQLite; no change needed)
# 4. Execute the whole script to create tables + insert data
cur.executescript(sql_text)
print("Database loaded!")
```

Which agents generate the most revenue?

Visualization: Bar chart – total revenue by agent

Python code:

```
q1_sql = """
SELECT
   a.AgentID,
   a.AgtFirstName || ' ' || a.AgtLastName AS agent_name,
   COUNT(e.EngagementNumber)             AS num_gigs,
   SUM(e.ContractPrice)                  AS total_revenue,
   AVG(e.ContractPrice)                  AS avg_contract
FROM Engagements e
JOIN Agents a
 ON e.AgentID = a.AgentID
GROUP BY a.AgentID, agent_name
ORDER BY total_revenue DESC;
"""
df_q1 = pd.read_sql_query(q1_sql, conn)
plt.figure(figsize=(8, 5))
plt.bar(df_q1["agent_name"], df_q1["total_revenue"])
plt.xticks(rotation=45, ha="right")
plt.ylabel("Total Revenue")
plt.title("Q1 – Revenue by Agent")
plt.tight_layout()
plt.show()
```

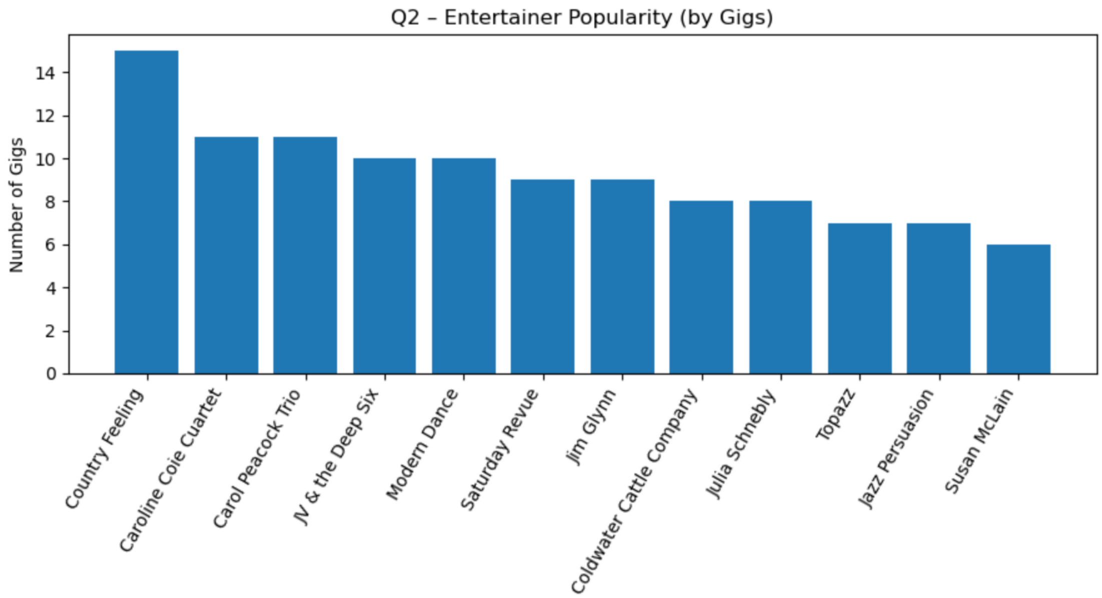


Which entertainers are the most popular and profitable?

Visualization: Bar – number of gigs per entertainer

Python code:

```
q2_sql = """
SELECT
    en.EntertainerID,
    en.EntStageName,
    COUNT(e.EngagementNumber) AS num_gigs,
    SUM(e.ContractPrice)      AS total_revenue,
    AVG(e.ContractPrice)      AS avg_contract
FROM Engagements e
JOIN Entertainers en
  ON e.EntertainerID = en.EntertainerID
GROUP BY en.EntertainerID, en.EntStageName
ORDER BY num_gigs DESC, total_revenue DESC;
"""
df_q2 = pd.read_sql_query(q2_sql, conn)
# Popularity: number of gigs
plt.figure(figsize=(9, 5))
plt.bar(df_q2["EntStageName"], df_q2["num_gigs"])
plt.xticks(rotation=60, ha="right")
plt.ylabel("Number of Gigs")
plt.title("Q2 – Entertainer Popularity (by Gigs)")
plt.tight_layout()
plt.show()
```



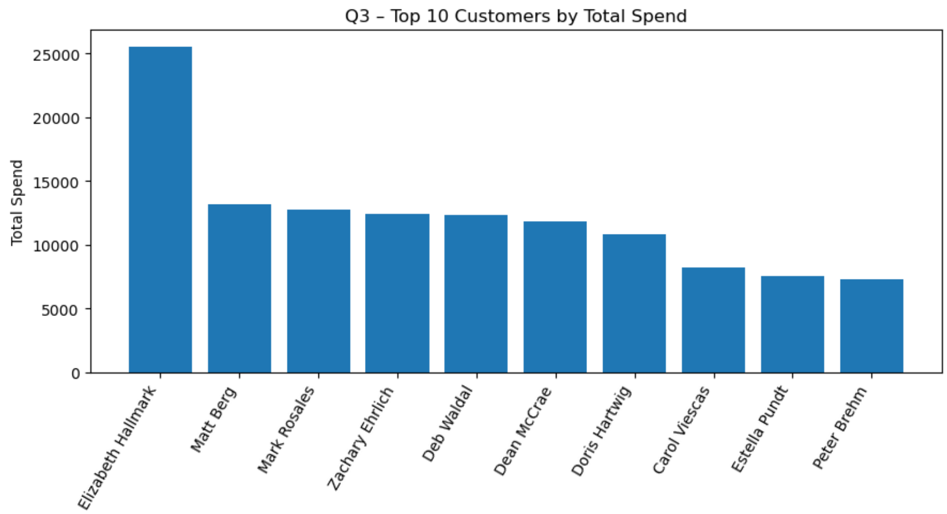
Which customers are the most valuable?

Visualization: Bar – top 10 customers by total spend

Python code:

```
q3_sql = """
SELECT
    c.CustomerID,
    c.CustFirstName || ' ' || c.CustLastName AS customer_name,
    c.CustCity,
    COUNT(e.EngagementNumber)              AS num_bookings,
    SUM(e.ContractPrice)                   AS total_spend,
    AVG(e.ContractPrice)                   AS avg_contract
FROM Engagements e
JOIN Customers c
  ON e.CustomerID = c.CustomerID
GROUP BY c.CustomerID, customer_name, c.CustCity
ORDER BY total_spend DESC;
"""
df_q3 = pd.read_sql_query(q3_sql, conn)
top_cust = df_q3.head(10)  # top 10 customers
plt.figure(figsize=(9, 5))
plt.bar(top_cust["customer_name"], top_cust["total_spend"])
plt.xticks(rotation=60, ha="right")
plt.ylabel("Total Spend")
plt.title("Q3 – Top 10 Customers by Total Spend")
```

```
plt.tight_layout()
plt.show()
```

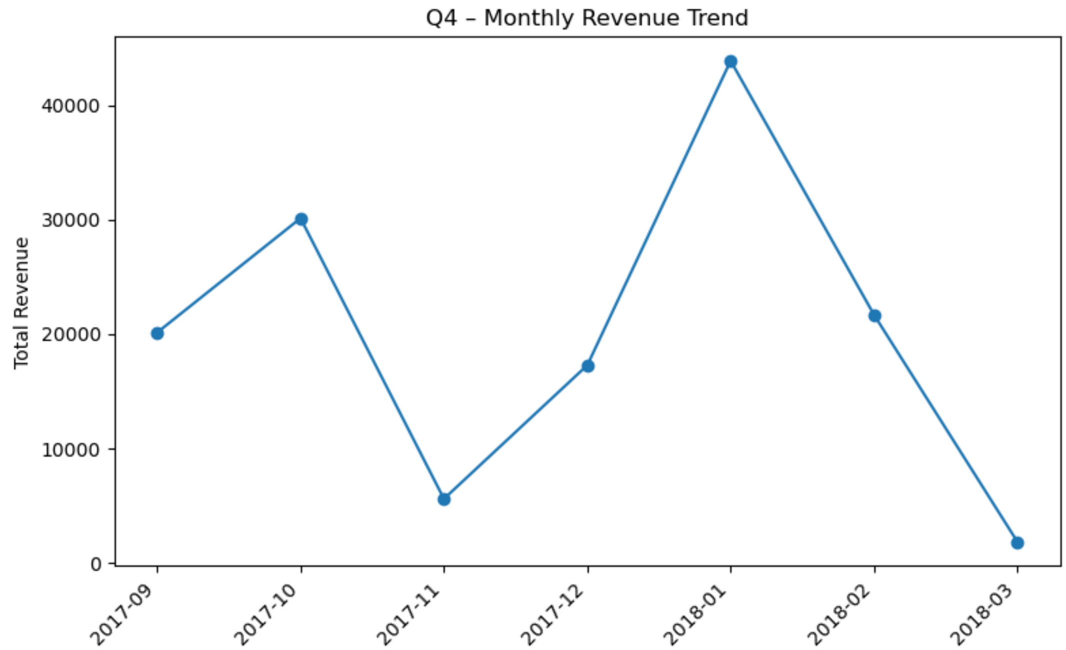


Is there seasonality in revenue (by month)?

Visualization: Line chart – monthly revenue trend

Python code:

```
q4_sql = """
SELECT
    substr(StartDate, 1, 7) AS month,   -- 'YYYY-MM'
    COUNT(EngagementNumber) AS num_gigs,
    SUM(ContractPrice)      AS total_revenue,
    AVG(ContractPrice)      AS avg_contract
FROM Engagements
GROUP BY substr(StartDate, 1, 7)
ORDER BY month;
"""
df_q4 = pd.read_sql_query(q4_sql, conn)
plt.figure(figsize=(8, 5))
plt.plot(df_q4["month"], df_q4["total_revenue"], marker="o")
plt.xticks(rotation=45, ha="right")
plt.ylabel("Total Revenue")
plt.title("Q4 – Monthly Revenue Trend")
plt.tight_layout()
plt.show()
```

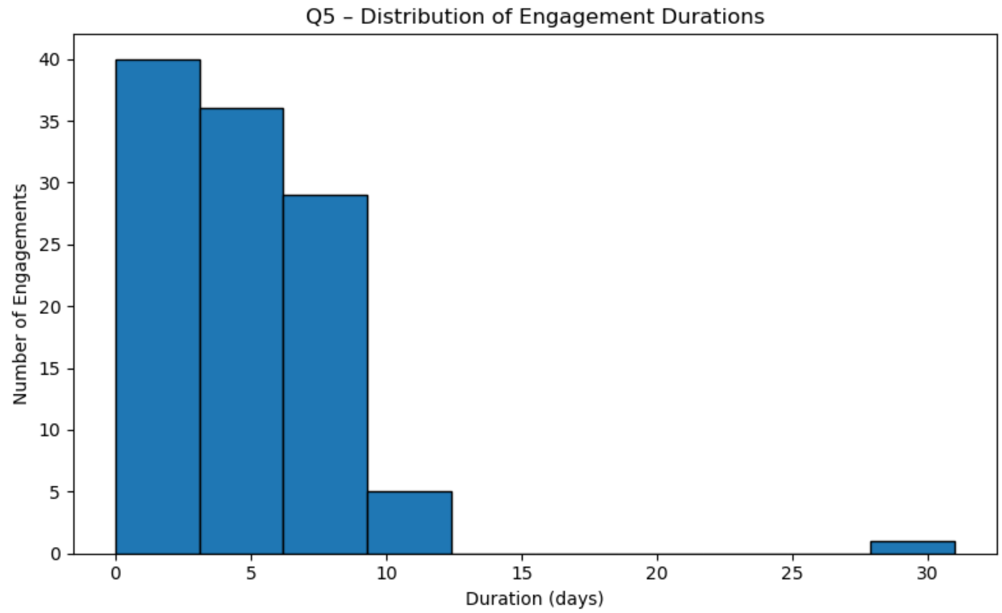


What is the typical duration of engagements?

Visualization: Histogram – distribution of durations in days

Python code:

```
q5_sql = """
SELECT
    EngagementNumber,
    StartDate,
    EndDate,
    julianday(EndDate) - julianday(StartDate) AS duration_days,
    ContractPrice
FROM Engagements;
"""
df_q5 = pd.read_sql_query(q5_sql, conn)
plt.figure(figsize=(8, 5))
plt.hist(df_q5["duration_days"], bins=10, edgecolor="black")
plt.xlabel("Duration (days)")
plt.ylabel("Number of Engagements")
plt.title("Q5 – Distribution of Engagement Durations")
plt.tight_layout()
plt.show()
```

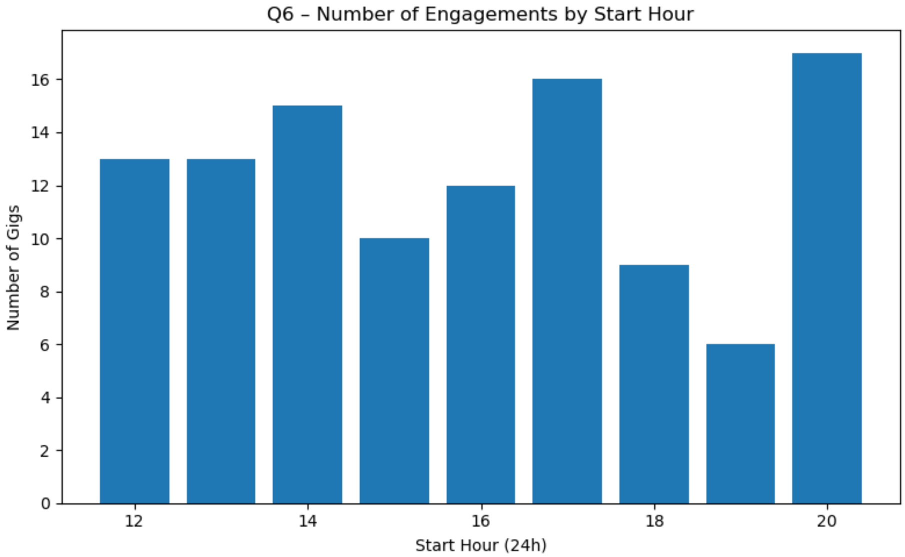


At what time of day do most engagements start?

Visualization: Bar – number of gigs by start hour (24h)

Python code:

```
q6_sql = """
SELECT
    CAST(strftime('%H', StartTime) AS INTEGER) AS start_hour,
    COUNT(*)                    AS num_gigs,
    SUM(ContractPrice)          AS total_revenue
FROM Engagements
GROUP BY CAST(strftime('%H', StartTime) AS INTEGER)
ORDER BY start_hour;
"""
df_q6 = pd.read_sql_query(q6_sql, conn)
plt.figure(figsize=(8, 5))
plt.bar(df_q6["start_hour"], df_q6["num_gigs"])
plt.xlabel("Start Hour (24h)")
plt.ylabel("Number of Gigs")
plt.title("Q6 – Number of Engagements by Start Hour")
plt.tight_layout()
plt.show()
```



Q6 – Number of Engagements by Start Hour
Number of Gigs
0
2
4
6
8
10
12
14
16
12
14
16
18
20
Start Hour (24h)

Compare and Contrast:

When comparing my SQL analysis to the work ChatGPT produced, several differences emerged in both process and analytical depth. Our approach focused on answering the business questions directly with clean, readable SQL and concise explanations. We identified the top booking customers, summarized the most common musical preferences, and analyzed high-value customers' preferred genres using straightforward joins, grouping logic, and clear CTE structures. A key feature of our work was being able to come up with more relevant questions to ask that would be more representative of the analysis we wanted to cover. For example which customers have made the most bookings and what are the most common musical preferences.

ChatGPT's work, however, was more structured and technically comprehensive. It incorporated more advanced SQL techniques, such as window functions and conditional aggregation, and applied more extensive data validation, including checks for missing customer records, preference ranking inconsistencies, and referential integrity across tables. ChatGPT also explored deeper analytical layers, like segmenting customers by value tiers, identifying co-occurring musical preferences, and comparing style distributions across different customer groups. Something unique about ChatGPT is that it was able to come up with other questions to explore that we hadn't originally thought of and was able to conduct analysis rather quickly. While ChatGPT delivered broader and more sophisticated analysis, our work provided focused insights that were easier to interpret and more directly tied to TuneWorks' business objectives. In essence, ChatGPT excelled in depth, technical rigor, and system-level completeness, while our work excelled in clarity, simplicity, and generating directly actionable findings for TuneWork's leadership. This EDA analysis was overall important because it demonstrated how both approaches have different strengths and how ChatGPT could be used to strengthen our findings we already made. An aspect that ChatGPT isn't able to replace, however, is the ability to come up with relevant recommendations and takeaways as we are able to factor into things that ChatGPT might now think of when it comes to dealing with a real company.

Example Results Comparison:

A. Customer Booking Volume

Your approach

Identified top customers based purely on engagement count.

Used:

RIGHT JOIN Customers c ON e.customerid = c.customerid

Highlighted high-value, mid-tier, and long-tail customers.

Provided practical business implications.

ChatGPT's approach:

Use LEFT JOIN + explicit NULL handling (more standard).

Add percentage-of-total bookings to quantify concentration.

Compute repeat rate, average time between bookings, or customer lifetime value.

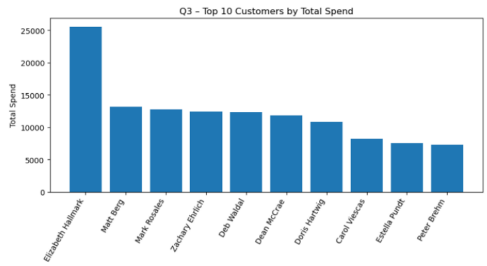
Create segments automatically (e.g., high-value, medium-value, one-and-done).
Example ChatGPT enhancement:
SUM(CASE WHEN total_bookings >= 8 THEN 1 END) AS high_value_count

B. Most Common Musical Preferences
Your approach:
Simple, clean frequency count of style preferences.
Query is valid and interpretable for business teams.

ChatGPT's approach:
Use preference ranking (preferenceseq) to weight preferences.
Identify co-occurring styles among customers.
Provide distribution charts or clusters of customers by taste.
Detect genre gaps in entertainer supply vs. customer demand.

Powerpoint generated by ChatGPT

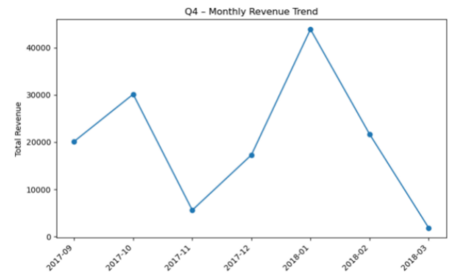
Q3 – Top 10 Customers by Total Spend



Elizabeth Hallmark stands out as a top-value customer, highlighting the importance of VIP-focused retention strategies. The remaining customers provide a stable long-tail revenue base.

5

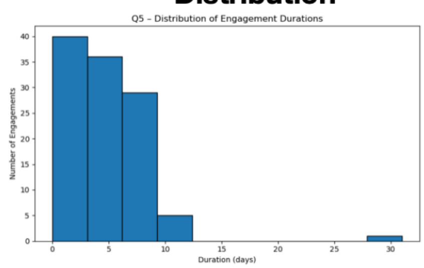
Q4 – Monthly Revenue Trend



January peaks sharply, driven by holiday and New Year events. November dips significantly, revealing potential for seasonal promotional opportunities.

6

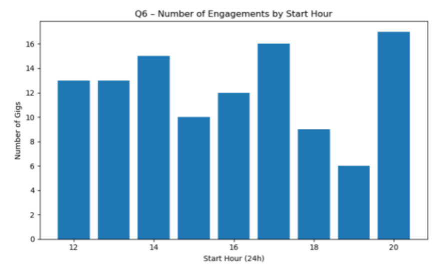
Q5 – Engagement Duration Distribution



Most engagements last 2–8 days, with rare long-duration events. This suggests flexible, short performance cycles tailored to client needs.

7

Q6 – Number of Engagements by Start Hour



Engagements cluster heavily between 12:00–20:00, with a peak at 20:00. This confirms high demand for evening entertainment sessions.

8

Conclusion

- • Revenue is concentrated among top agents and entertainers.
- • Strong seasonality creates opportunities for targeted campaigns.
- • VIP customer identification reveals valuable long-term clients.
- • Engagement durations show short-term performance preferences.
- • Evening hours dominate booking trends, aligning with entertainment norms.

- Overall, the EDA provides clear guidance for operational planning, marketing strategy, and resource allocation.

9